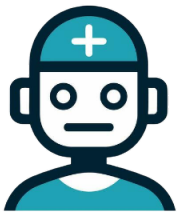
UNIVERSITY OF BIRMINGHAM DUBAI

برنامج الذكاء الاصطناعي
THE ARTIFICIAL INTELLIGENCE PROGRAM



# Securing Health Data for Clinical Agent

## Overview of Data Features:

| Field(s) | Meaning |
|---|---|
| encounter_id, patient_id | Unique stable record/person tokens |
| patient_name, date_of_birth, phone, email, street_address | Direct identifiers |
| age, gender | May be predictive |
| postal_code, region, city, neighborhood | Geographic specificity |
| occupation | Person's job |
| household_children, other_dependents, lives_alone, housing_status, caregiver_relation | Social/care indicators |
| preferred_language | Access-to-care context |
| admission_date, discharge_date | Exact date admitted to and released from hospital |
| diagnosis, medication | Sensitive clinical features |
| prior_admissions_6m | Number of admissions in the last 6 months |
| length_of_stay_days | Number of days in hospital during this admission |
| abnormal_lab_flag | Did they have abnormal lab result? |
| missed_appointments_3m | How many appointments did they miss in the last 3 months? |
| transport_barrier, mobility_limitation, medication_complexity | Any special needs |
| discharge_note | Free-text reports summarising their diagnosis, treatment, medications, and recommended follow-up care |
| follow_up_7d | Fabricated binary training label, 1 = priority care-coordinator review |

## Techniques to Secure Data:

| Technique | Meaning |
|---|---|
| Data Minimisation | **Drop** contact data, unnecessary details; retains least needed for task |
| Text Redaction | **Mask** names, addresses, dates and *contextual* clues |
| Pseudonymisation | Substitute stable identifiers but preserve linkability; **(not anonymisation)** |
| Generalisation | Convert exact age to bands; **coarsen regions** and dates; may reduce utility |
| Suppression | **Withhold rare** categories or small-group values |
| k-anonymity | Make selected **quasi-identifier groups** size >= k |
| Perturbation | Small **random noise** to numeric fields |
| Differential privacy | Requires calibrated mechanism and **privacy budget** |
| Synthetic data | Add **artificial samples** |